Circles

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Inequalities in one triangle

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

2 [Foundational] • Target: Inequalities in one triangle

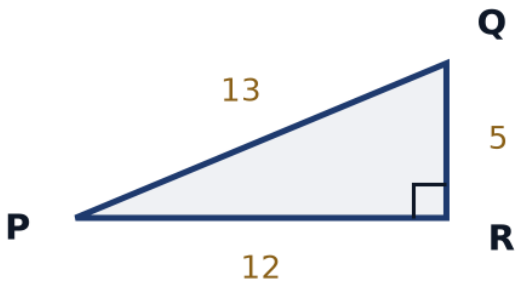
On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

3 [Foundational] • Target: Inequalities in one triangle

In ΔABC studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

4 [Foundational] • Target: Inequalities in one triangle

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.



5 [Foundational] • Target: Inequalities in one triangle

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

6 [Foundational] • Target: Inequalities in one triangle

In ΔABC studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

7 [Foundational] • Target: Inequalities in one triangle

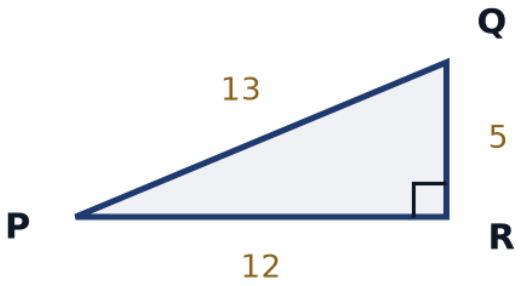
A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

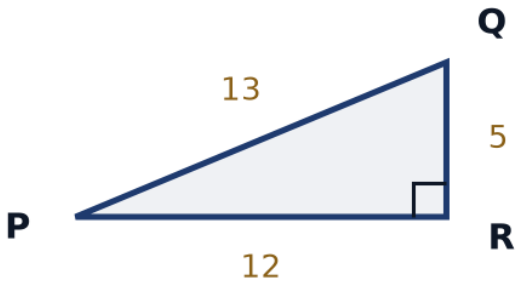
8 **[Intermediate]** • **Target: Inequalities in one triangle**

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



9 **[Intermediate]** • **Target: Inequalities in one triangle**

In ΔABC studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



10 **[Intermediate]** • **Target: Inequalities in one triangle**

A geometric figure scaled by factor $k = 5$ has an original side length of $7\ \text{cm}$. What is the corresponding image length? Show your complete algebraic work.

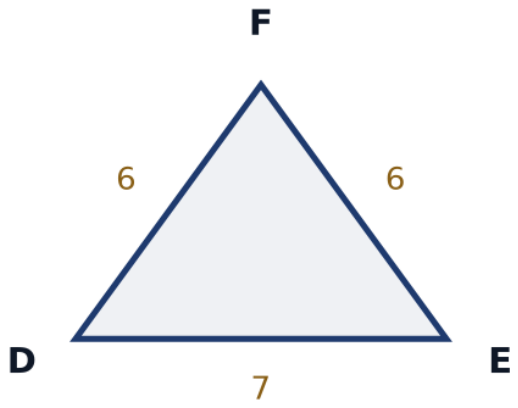
11 **[Intermediate]** • **Target: Inequalities in one triangle**

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

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12 [Intermediate] • Target: The triangle inequality

In ΔABC studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



13 [Intermediate] • Target: The triangle inequality

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

14 [Intermediate] • Target: The triangle inequality

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

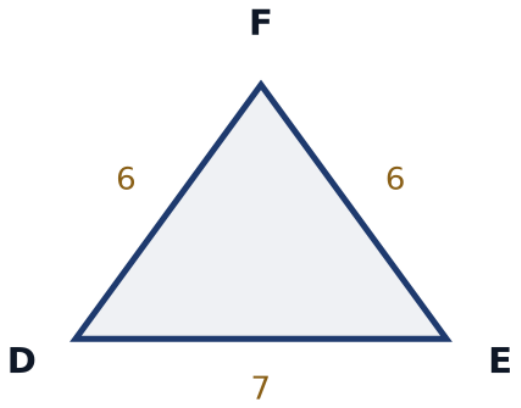
15 [Challenging] • Target: The triangle inequality • Cross-Lesson: 11.01 Probability

In ΔABC studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

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16 [Challenging] • Target: The triangle inequality • Cross-Lesson: 2.01 Lines & Angles

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.



17 [Challenging] • Target: The triangle inequality • Cross-Lesson: 3.01 Triangles & Congruence

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

18 [Challenging] • Target: The triangle inequality • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

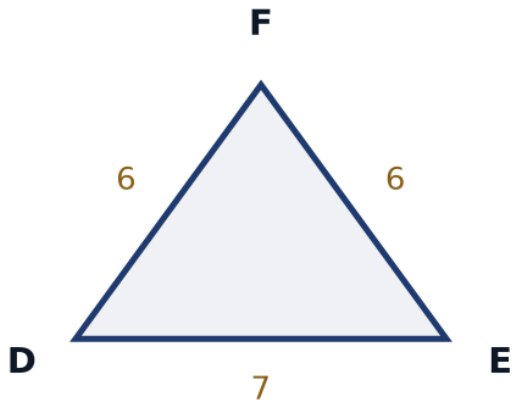
In ΔABC studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

19 [Challenging] • Target: The triangle inequality • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

20 [Challenging] • Target: The triangle inequality • Cross-Lesson: 6.01 Polygons & Quadrilaterals

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

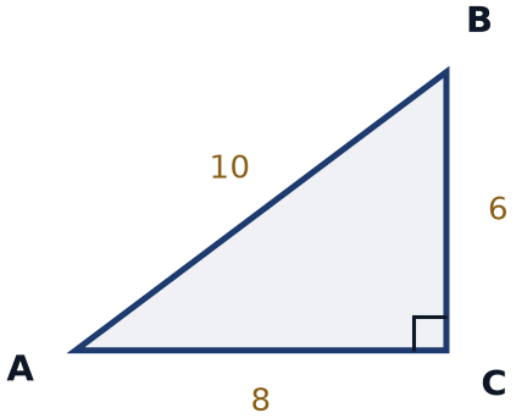


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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Angle bisectors and perpendicular bisectors

In ΔABC studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 94°

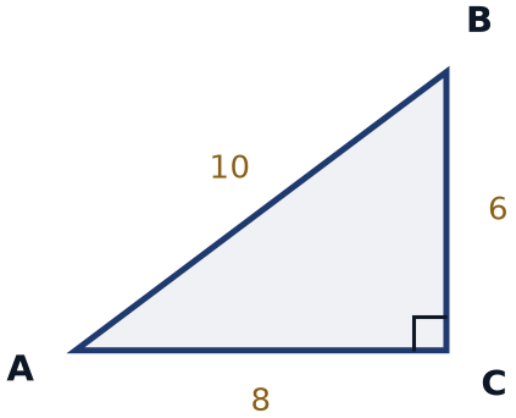
b. 104°

c. 86°

d. 139°

22 [Foundational] • Target: Angle bisectors and perpendicular bisectors

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?



a. 12 cm

b. 15 cm

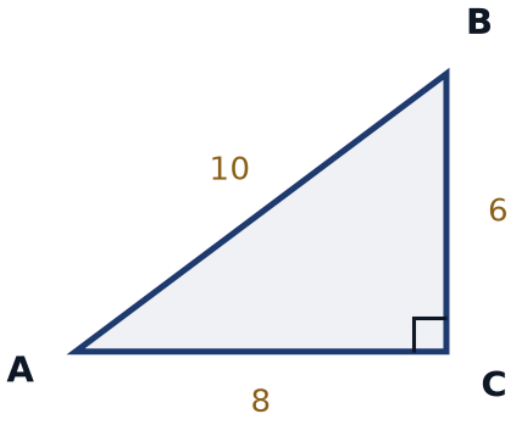
c. 20 cm

d. 45 cm

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23 [Foundational] • Target: Angle bisectors and perpendicular bisectors

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?



a. 8

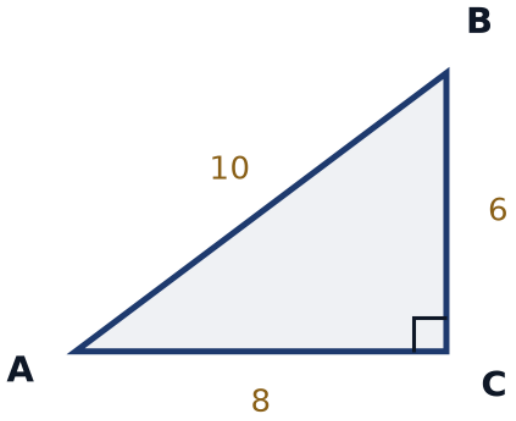
b. 12

c. 10

d. 12

24 [Foundational] • Target: Angle bisectors and perpendicular bisectors

In ΔABC studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 73°

b. 83°

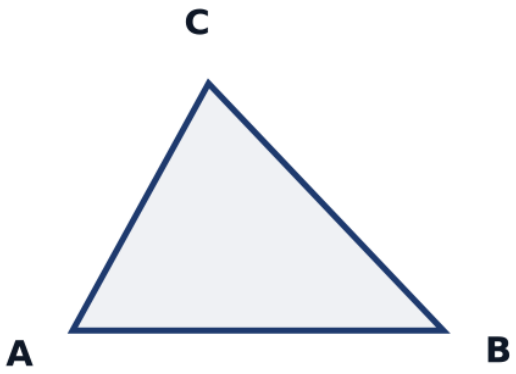
c. 65°

d. 139°

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25 [Foundational] • Target: Altitudes, medians, and bisectors of triangles

In ΔABC studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 93°

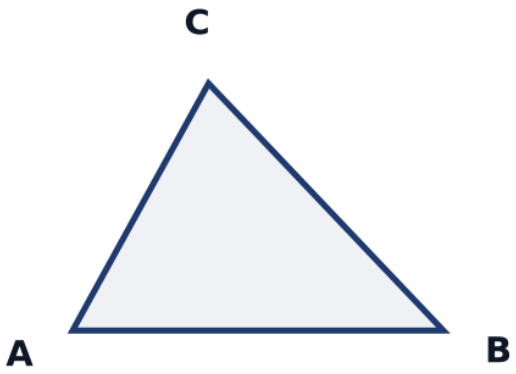
b. 103°

c. 85°

d. 138°

26 [Foundational] • Target: Altitudes, medians, and bisectors of triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?



a. 15 cm

b. 18 cm

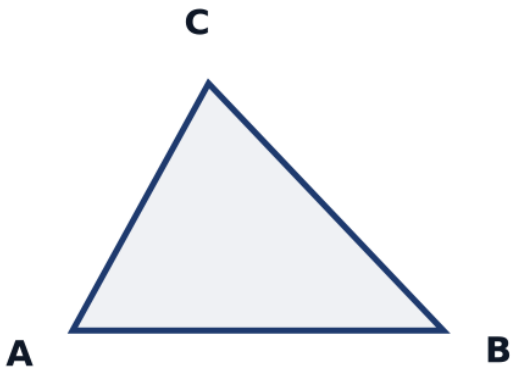
c. 24 cm

d. 54 cm

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27 [Foundational] • Target: Altitudes, medians, and bisectors of triangles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$?



a. 11

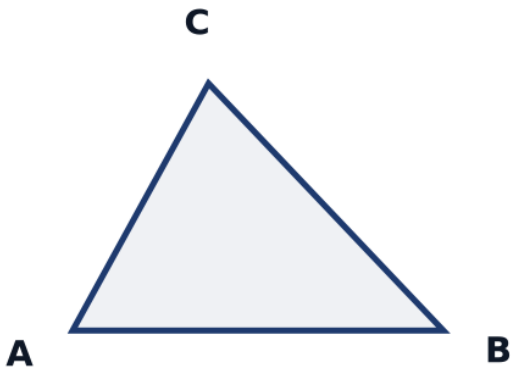
b. 16

c. 14

d. 17

28 [Foundational] • Target: Altitudes, medians, and bisectors of triangles

In ΔABC studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 72°

b. 82°

c. 64°

d. 138°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Angle bisectors and perpendicular bisectors

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm

c. 15 cm

d. 20 cm

30 [Intermediate] • Target: Angle bisectors and perpendicular bisectors

In ΔABC studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 87°

b. 97°

c. 79°

d. 139°

31 [Intermediate] • Target: Angle bisectors and perpendicular bisectors

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm

c. 25 cm

d. 80 cm

32 [Intermediate] • Target: Altitudes, medians, and bisectors of triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

33 [Intermediate] • Target: Altitudes, medians, and bisectors of triangles

In ΔABC studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 86°

b. 96°

c. 78°

d. 138°

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34 [Intermediate] • Target: Altitudes, medians, and bisectors of triangles

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm

c. 30 cm

d. 96 cm

35 [Intermediate] • Target: Inequalities in one triangle

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

36 [Intermediate] • Target: Inequalities in one triangle

In ΔABC studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 85°

b. 95°

c. 77°

d. 137°

37 [Intermediate] • Target: Inequalities in one triangle

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm

c. 35 cm

d. 112 cm

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: The triangle inequality • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm

c. 24 cm

d. 32 cm

39 [Challenging] • Target: The triangle inequality • Cross-Lesson: 5.01 Right Triangles & Trigonometry

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 12

b. 19

c. 17

d. 22

40 [Challenging] • Target: The triangle inequality • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm

c. 40 cm

d. 128 cm

41 [Challenging] • Target: Extension: Geometric mean • Cross-Lesson: 8.01 Area, Surface Area & Volume

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm

c. 12 cm

d. 16 cm

42 [Challenging] • Target: Extension: Geometric mean • Cross-Lesson: 9.01 Transformations & Symmetry

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 10

b. 14

c. 12

d. 14

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43 [Challenging] • Target: Extension: Geometric mean • Cross-Lesson: 1.01 Foundations & Reasoning

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

44 [Challenging] • Target: Dividing a segment into a given ratio • Cross-Lesson: 10.01 Coordinate Geometry

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

45 [Challenging] • Target: Dividing a segment into a given ratio • Cross-Lesson: 11.01 Probability

For the model $f(x) = 3x + 9$ in dividing a segment into a given ratio, evaluate $f(4)$.

a. 21

b. 24

c. 12

d. 25

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